

Imitation Learning and Reinforcement Learning

Deep Learning Lab Course 2026

Ajaipaul Singh Bajwa
Matriculation Number: 6105553

1 Imitation Learning

A training dataset of almost 30000 steps was collected by the expert. A CNN based behavioral cloning agent was trained on expert driving data collected from the CarRacing-v3 environment. The network takes grayscale image frames as input and predicts driving actions.

1.1 Hyperparameters

The following hyperparameters were used to train the Behavioural Cloning agent:

Table 1: Hyperparameters used for Imitation Learning

Mini-Batches	Batch Size	Learning Rate	Optimizer	Loss	History Length	Training Samples	Network
3000	256	1×10^{-4}	Adam	Cross-Entropy	8	30000 steps	CNN

1.2 Results over 15 Episodes

The agent was evaluated over 15 episodes with different history lengths. Results are shown in Table 2 and training curves are provided in Figure 1 in the Appendix.

Table 2: Imitation Learning – results across different history lengths

History N	Mean Score	Std Deviation
1	52.80	73.34
3	286.68	123.99
5	286.84	223.67
8	485.89	231.80

The results show that the agent’s performance consistently increased as the history length was increased, suggesting that longer temporal context helps the agent make better driving decisions.

2 Reinforcement Learning

DQN agent was first trained on CartPole using an MLP network, then extended to CarRacing-v3 using a CNN architecture. Both agents use epsilon-greedy exploration, experience replay, and a target network to stabilize training. Adam was used as the optimizer along with MSE loss.

2.1 Hyperparameters

The following hyperparameters were used to train the CartPole experiment:

Table 3: Hyperparameters used for CartPole Experiment

Network	History Len.	ϵ Start	ϵ Decay	Learning Rate	Discount γ	Episodes	Batch Size
FC (64N)	1	0.2	0.995	1×10^{-4}	0.95	360	64

The following hyperparameters were used to train the CarRacing experiment:

Table 4: Hyperparameters used for CarRacing Experiment

Network	History Len.	ϵ Start	ϵ Decay	Learning Rate	Discount γ	Max Timesteps	Episodes	Batch Size
CNN (3C+2FC)	1	0.9	0.995	1×10^{-4}	0.95	1000	1000	64

2.2 Results over 15 Episodes

The agents were evaluated over 15 episodes after training. Results are shown in Tables 5 and 6 and training curves are provided in Figure 2 in the Appendix.

Table 5: Reinforcement Learning (CartPole) – evaluation results over 15 episodes

Episode	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Mean	Std
Reward	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500.00	0.00

Table 6: Reinforcement Learning (CarRacing) – evaluation results over 15 episodes

Episode	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Mean	Std
Reward	848.72	893.06	856.52	864.17	792.98	858.90	846.04	860.71	881.13	872.05	552.32	888.10	918.90	859.87	851.61	843.01	82.14

The DQN agent performed significantly better than the Behavioural Cloning agent on CarRacing, achieving a mean reward of 843.01 ($\sigma = 82.14$) compared to 485.89 ($\sigma = 231.80$) for the BC agent.

AI Tool Usage Disclosure

AI tools were used for the following purposes:

- Understanding the concepts behind Reinforcement and Imitation Learning.

- Understanding the given codebase, the variables used, and how they affect the model.
- Debugging and code cleanup.
- Proofreading this report and fixing any mistakes.

Appendix



Figure 1: Imitation Learning – training curves

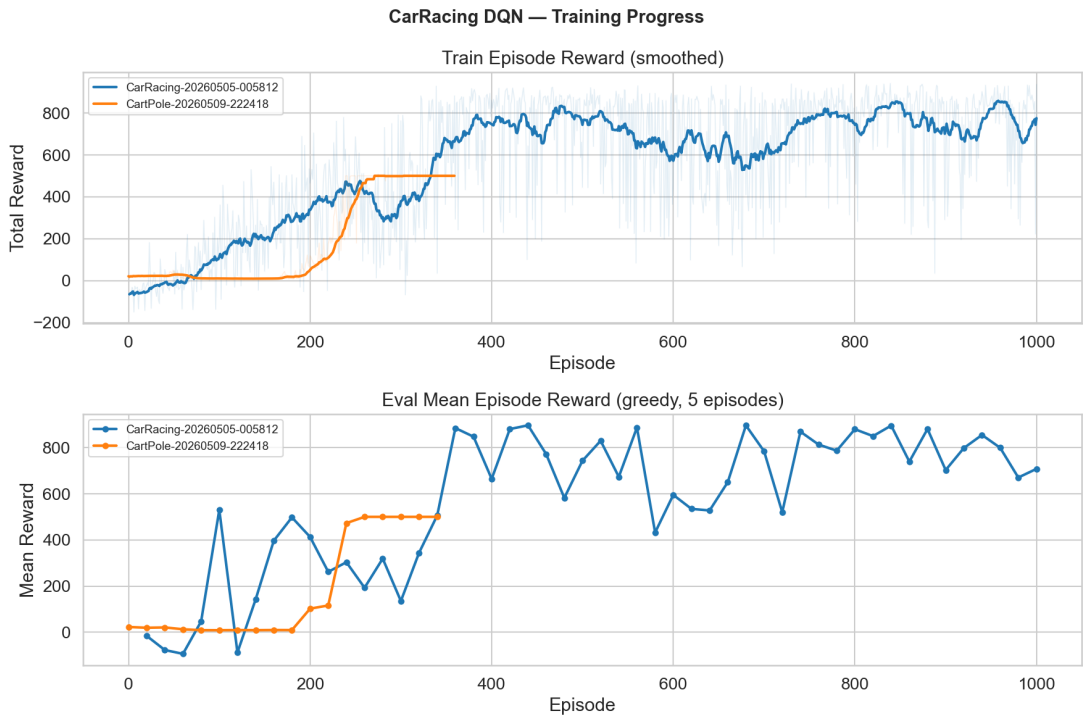


Figure 2: Reinforcement Learning – training curves